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OntoEval: an Automated Ontology Evaluation System

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ABSTRACT

Developing semantically-aware web services requires comprehensive and accurate ontologies. Evaluating an existing ontology or adapting it is a labor-intensive and complex task for which no automated tools exist. Nevertheless, in this paper we propose a tool that aims at making this vision come true, i.e., we present a tool for the automated evaluation of ontologies that allows one to rapidly assess an ontology's coverage of a domain and identify specific problems in the ontology's structure. The tool evaluates the domain coverage and correctness of parent-child relations of a given ontology based on domain information derived from a text corpus representing the domain. The tool provides both overall statistics and detailed analysis of sub-graphs of the ontology. In the demo, we show how these features can be used for the iterative improvement of an ontology.

CCS CONCEPTS

• **Information systems** → *Web Ontology Language (OWL)*; *Web services*; **Information integration**.

KEYWORDS

ontology, natural language processing, BERT, knowledge engineering

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1 INTRODUCTION

Ontologies are an essential component of the Semantic Web, as they provide a shared understanding of the meaning and relationships of the concepts and terms used in a particular domain. Ontologies further enable semantically aware data integration services (OBDI) [8]. One such example is the Oceanographic Data Integration Initiative (ODINI[1]), which provides oceanographic researchers with the ability to automatically integrate scientific data previously collected for other research projects [16] for the purpose of meta-analysis or longitudinal evaluation [15]. In OBDI, ontologies serve an important role in unifying data elements from different datasets under the same concepts despite each dataset having a different schema and potentially using slightly different terminology in its attributes to describe similar concepts. Furthermore, the semantic relation between the two schemas' attributes can be identified using the ontology. For example, in the domain of oceanography, *Phosphate*[2] refers to the amount of Phosphorous-based macro-nutrients in seawater, such as *iron phosphate*[3] or *sodium phosphate*[4]. Thus, if a researcher is looking to unify data about the nutrient content of water from different datasets, where some contain data about *iron phosphate* and the other about *sodium phosphate*, the semantic relation between the two can be determined from the oceanographic ontology.

However, the construction of an ontology remains a challenge, as it is often constructed manually by domain experts [1] requiring many hours of work by resources with limited availability. When required to modify an ontology or to assess its fit, one needs to evaluate the correctness of the ontology and its coverage of the domain. Existing approaches of automated ontology evaluation are limited in two aspects: the first is concept extraction from the domain; existing methodologies only consider the use of Part-of-Speech (POS) models, which can only determine single-word terms. Multi-word phrases, such as *Air Temperature*, will be split into separate concepts (*Air*, *Temperature*). Moreover, existing approaches in which ontological relationships are evaluated with respect to a domain only consider their semantic relations with respect to external knowledge bases such as thesauruses, dictionaries, and vocabularies, all of which are general-purpose and are not representative of the domain. These gaps led us to develop OntoEval. Our tool uses a novel evaluation framework to measure the accuracy and completeness of ontologies. It employs a domain-specific Named-Entity-Recognition (NER) model for detecting multi-word phrases and labeling their types and a pre-trained BERT [5] model to determine the similarity and relatedness of concepts. This enables us to assess the relevance and correctness of an ontology in relation to a specific domain.

2 BACKGROUND AND RELATED WORK

As defined by Gruber [9], "An ontology is an explicit specification of a conceptualization." The representation is made through a collection of concepts and relations between them. Ontologies are a crucial part of web semantics and information systems as they capture a representation of the domain such that it can be interpreted by machines required in tasks such as information retrieval [17], data integration [7], and knowledge alignment [11]. When

[1] https://odini.net/, retrieved January 15th, 2023

[2] https://www.ebi.ac.uk/chebi/chebiOntology.do?chebiId=CHEBI:26020
[3] https://www.ebi.ac.uk/chebi/chebiOntology.do?chebiId=CHEBI:131371
[4] https://www.ebi.ac.uk/chebi/chebiOntology.do?chebiId=CHEBI:37586

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evaluating an ontology, we use the term *concept family*, to represent an entity comprised of a parent and a set of child concepts with a direct hierarchical relation to the parent concept.

Raad et al. [14] identified four methods and seven criteria for evaluating ontologies. Using this framework, we define our evaluation as a *corpus-based method*, as it extracts terms from a corpus to determine the fit of the evaluated ontology to the domain represented by the corpus. Corpus-based approaches are more robust than those that rely on structured knowledge sources, such as a vocabulary or taxonomy, which is also limited as it requires much of the same high-cost resources to construct. Its existence otherwise obviates the need for an ontology. Of the seven criteria defined by Raad et al. (accuracy, completeness, conciseness, adaptability, clarity, computational efficiency, consistency), our method focuses on accuracy, completeness, and conciseness.

Previous work on corpus-based methods [3, 6] have been limited to single-word nouns and used general-purpose dictionaries to determine synonyms and related concepts that do not necessarily reflect the true domain hierarchy.

3 AUTOMATED EVALUATION METHOD

Here, we describe our automated approach for evaluating an ontology with respect to a domain of interest. We evaluate the overall ontology for completeness (coverage), conciseness, and correctness (semantic relation coherence). While the approach is domain-agnostic, the proposed demo will be loaded with a BERT model pre-trained on the oceanographic domain.

Fig. 1 showcases the evaluation pipeline in which a collection of text documents or text corpus (Section 3.1) is used to train a named-entity-recognition (NER) model (Section 3.2) and pre-train a language model (Section 3.5). Using the NER model, domain concepts are extracted from the text (Section 3.3) and are then matched up with the concepts in the ontology (Section 3.4), from which a derived sub-ontology is generated. Next, using the pre-trained language model, an evaluation (Section 3.6) takes place, generating a set of scores that reflect the applicability of the ontology in that domain.

3.1 Document Collection

The corpus is assumed to encompass the domain knowledge. Large text collections are routinely used in this capacity for tasks ranging from training domain-specific language models [10] to guiding automated literature surveys [2]. Moreover, the alternative, relying on a few domain experts, is both costly and arguably less representative.

3.2 Domain Specific Named Entity Recognition Model

A typical NER model is capable of identifying general entities in text, such as human names, locations, or years. But in order to extract concepts relevant to the domain, for example, *temperature*, one must train a custom NER model [13].

3.3 Domain Concepts

In order to evaluate the ontology with respect to a given domain, we must first list the domain concepts. We accomplish this by using the NER model described above in Section 3.2, in addition to a general-purpose PoS noun extractor. Unlike part-of-speech (PoS) tagging, which determines the grammatical composition of tokens, a NER model can not only detect multi-word phrases but also semantically label them. For example, *Organization* or *Measured Variable*. After extraction, low-occurrence concepts are removed.

3.4 Ontology Subset Construction

To reduce clutter caused by unrelated concepts throughout the evaluation phase, a domain-relevant subset of the ontology is derived. This is done by matching all concepts from the evaluated ontology to the previously extracted domain concepts. This phase is done by processing concept labels and includes lower-casing and lemmatization. After applying this process to both the domain and ontological concepts, an exact match search for overlapping concepts can begin. We refer to these overlapping concepts as *Shared Concepts*. Using the shared concepts we traverse the ontology's hierarchy such that every ancestor in the hierarchy of a shared concept is included, as well as every direct child of that concept.

3.5 Language Model Pre-training

Using the same textual corpus data collected, we pre-train a BERT [5] language model using its standard unsupervised pre-training method. We later utilize this model's embedding layer to encode evaluated concepts.

3.6 Evaluation of Completeness and Correctness

In order to determine the completeness and correctness of the ontology subset as defined in Section 3.4, we compute three metrics: *Ontology Relevance*: $\frac{\#\text{shared concepts}}{\#\text{ontology concepts}}$, *Sub-Ontology Relevance*: $\frac{\#\text{shared concepts}}{\#\text{sub-ontology concepts}}$, and *Domain Coverage*: $\frac{\#\text{shared concepts}}{\#\text{domain concepts}}$ based upon the shared concepts found between the ontology and the domain concepts extracted from the corpus. To evaluate the correctness of the semantic relationships between ontological concepts, we define the following measures, which utilize concept embedding using the domain-adapted language model.

Child Similarity Score - CSS is the mean of mean cosine similarities of every sibling in a concept family:

$$CSS(CF) = \frac{1}{M}\sum_{i=1}^{M}[\frac{1}{M-1}\sum_{j=1}^{M} similarity(C_i, C_j)_{\text{ if } i \neq j}] \quad (1)$$

Parent Similarity Score - PSS is the mean cosine similarity between the parent C_p and each of its M direct child concepts:

$$PSS(CF) = \frac{1}{M}\sum_{i=1}^{M} similarity(C_i, C_p) \quad (2)$$

Parent Difference Agreement - PDA uses the standard deviation of the similarity between the parent and its direct children to reflect consistency. We can interpret this value as the amount of agreement between the siblings towards the parent:

$$PDA(CF) = 1 - \sqrt{\frac{1}{M-1}\sum_{i=1}^{M}[similarity(C_i, C_p) - PSS(CF)]^2} \quad (3)$$

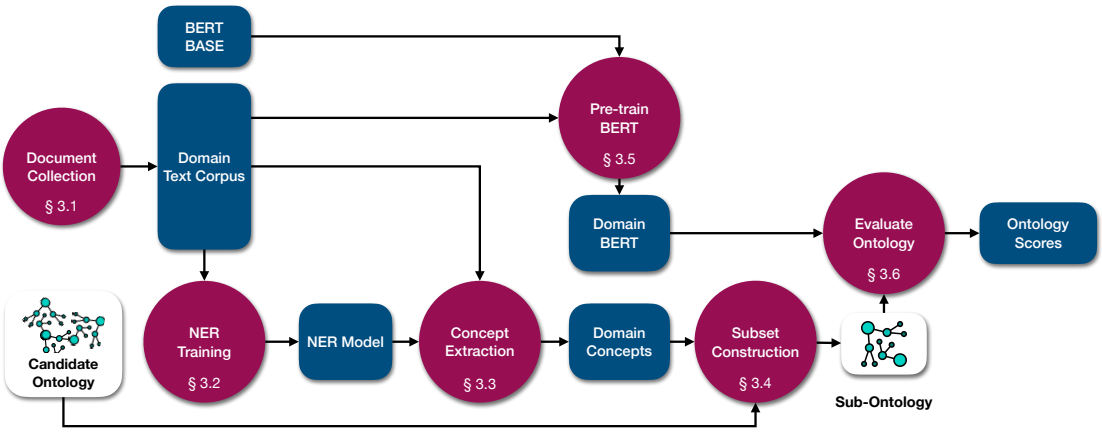


Figure (1) Ontology evaluation pipeline

Using the defined measures, we iterate over all concept families.

4 DEMONSTRATION INTENTION

We now present the system that we will demonstrate and evaluation examples on an Oceanographic ontology. The domain model used here was created using a corpus of 10,000 oceanographic academic papers and a NER model trained on it from previous research [15]. Using the NER model and a general-purpose PoS tagger [12], we extracted concepts from the text corpus. This phase generated 36,000 unique concepts. After filtering for frequency and term length, 12,000 concepts remained. We also pre-trained the language model BERT[5] [5] using the same corpus. With the domain concepts and a given ontology, we create two sets: one containing all the unique domain concepts and another containing all unique ontology concepts. The creation of the shared concept set and sub-ontology is done with RDFLib[6]. A SPARQL[7] query the subset ontology is used to extract the concept families. We then iterate over them to compute the measures defined above. Cosine similarity determines the semantic similarity between concepts as calculated between the domain-encoded embedding of their label.

4.1 Utilizing OntoEval for Ontology Refinement

Here, we demonstrate how OntoEval can be used to extend and refine an ontology. Consider a use case where we would wish to create a comprehensive data integration ontology on the basis of ENVO [4]. When evaluating the coverage of ENVO with OntoEval, the top-level measures are shown in Figure 2. These suggest that there is low coverage (0.05) of the target domain. To identify which missing concepts may be candidates for inclusion, one can use the detailed coverage result analysis shown in Figure 2. Thus, one could add concepts from this list to a new version of the ENVO ontology and rerun the evaluation.

ENVO also presents with a relatively low PDA. In order to gain better insight into the type of problems the model has identified, we can zoom into a specific concept family such as the *Environmental Condition* concept family (Figure 3. It seems that the *environmental variability* concept has a low PDA with the other concepts in this concept family. We could consider splitting this family, excluding this concept, or refining the type of relation used between the concepts here. To preview some of these ideas, we can use the CSS matrix screen (Figure 4). On the same screen, we can see the effects of adding a concept to this concept family. Here, we attempt to add a missing environmental condition (continental).

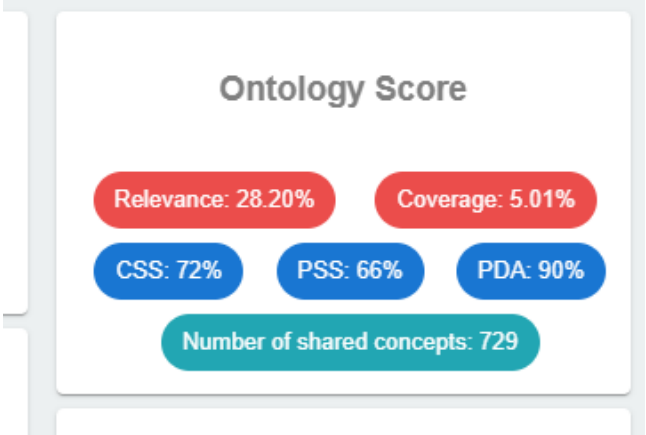


Figure (2) Top-level evaluation of the ENVO ontology

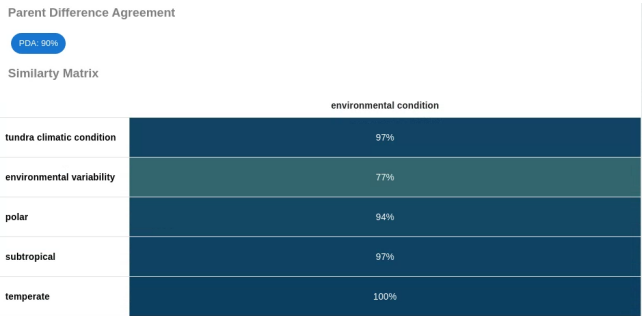


Figure (3) PDA values of child concepts of *Environmental Condition* from ENVO

[5]BERT Base, Transformers 4.17.0, https://huggingface.co/, accessed June 6th, 2022
[6]RDFLib https://rdflib.readthedocs.io/en/stable/
[7]SPARQL https://www.w3.org/TR/rdf-sparql-query/

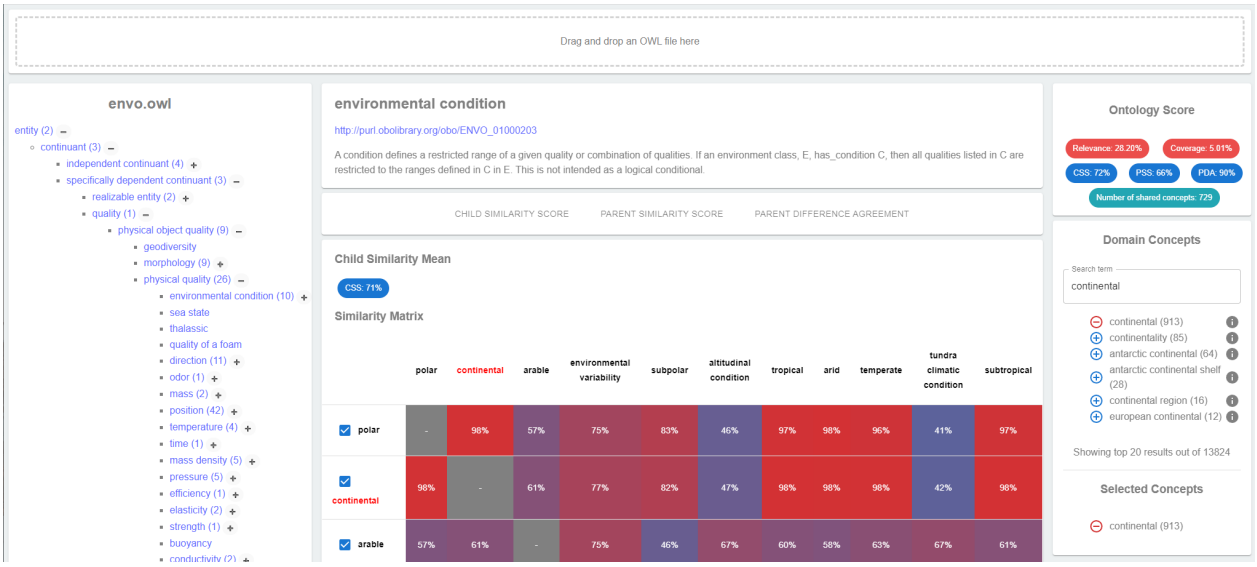


Figure (4) CSS matrix of child concepts of *Environmental Condition* from ENVO

5 CONCLUSION

We demonstrate *OntoEval*, an automated ontology evaluation system designed to support the refinement of an ontology. *OntoEval* is based upon a representation of the domain of interest created by a large language model trained on the domain. *OntoEval* allows users to evaluate an ontology for its coverage and relevance, identify missing concepts, and explore concept families to identify mismatched concepts and ill-structured families. Additional details, including external validation of our model's performance, are available [18].

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